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$$2(3z^2 - z + 1) + i(z + 1) = 0$$

$$\Leftrightarrow 6z^2 - 2z + 2 + iz + i = 0$$

$$\Leftrightarrow 6z^2 + (-2 + i)z + (2 + i) = 0$$

$$D = (-2 + i)^2 - 4 \cdot 6 \cdot (2 + i)$$

$$(-2 + i)^2 = 4 - 4i + i^2 = 3 - 4i$$

$$4 \cdot 6 \cdot (2 + i) = 24(2 + i) = 48 + 24i$$

$$D = (3 - 4i) - (48 + 24i) = 45 - 28i$$

$$\begin{cases} x^2 - y^2 &= -45 \\ 2xy &= -28 \Rightarrow xy = -14 \Rightarrow x^2(-y^2) = -196 \end{cases}$$

$$\Rightarrow \lambda^2 + 45\lambda - 196 = 0$$

$$D = 45^2 + 4 \cdot 196 = 2025 + 784 = 2809$$

$$\lambda_1 = \frac{-45 + 53}{2} = 4 = x^2 \Rightarrow x = \pm 2$$

$$\lambda_2 = \frac{-45 - 53}{2} = -49 = -y^2 \Rightarrow y = \pm 7$$

$$\Rightarrow (x + yi) \in \{2 - 7i, -2 + 7i\}$$

$$z = \frac{-(-2 + i) \pm (2 - 7i)}{12}$$

$$z_1 = \frac{2 - i + 2 - 7i}{12} = \frac{4 - 8i}{12} = \frac{1}{3} - \frac{2}{3}i$$

$$z_2 = \frac{2 - i - 2 + 7i}{12} = \frac{6i}{12} = \frac{i}{2}$$